

The AI Act: Towards global effects ?

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Abstract

This paper examines the European Union's Artificial Intelligence Act (AIA) and its potential to become a global gold standard for trustworthy AI. The AIA aims to address ethical concerns and ensure responsible AI development. The paper highlights the need for AI guidelines, discusses the key features of the AIA, and explores controversies surrounding the legislation, particularly regarding newer generative AI technologies. It also discusses the ongoing discussions and potential for a "*Brussels effect*", where the AIA's standards influence global AI regulation. The paper concludes by emphasizing the importance of a regulation that sets guardrails for AI while promoting understanding and collaboration with the private sector.

1 Introduction

"On artificial intelligence, trust is a must, not a nice to have" proclaimed Margrethe Vestager - Executive Vice-President for a Europe fit for the Digital Age, during the public announcement of the European Union's (EU) first Artificial Intelligence Act (AIA) in 2021 (Sutcliffe (2022)). In a digitised era where artificial intelligence (AI) is increasingly integrated in smart products and services, the EU's proposed new set of rules on AI aims to ensure that the potential of the technology is harnessed in alignment with Europe's values (Feingold (2023)). Given the mounting ethical concerns as AI developments multiply (Pazzanese (2020)), the EU plans on adopting an ex-ante policymaking strategy to enforce appropriate ethical AI guidelines. Therefore, could the AI act become a global gold standard for trustworthy AI? This paper will argue that while the AI act has the potential to have a Brussels effect, known as the externalization of EU's standards (Bradford (2012)), it first needs to settle a relevant European standard despite controversies around the piece of legislation.

2 The need for AI guidelines

In the last decade, harmful cases of AI have proved the dangers of deploying the technology at scale and have as a consequence urged a response from policymakers. Already back in 2016, Microsoft's Twitter experiment called "*Taybot*" revealed the threats of developing and releasing an AI-powered account, as the chatbot autonomously tweeted racist, sexist and anti-Semitic content (Database (n.d.)). As stated

by experts, “*unmanaged AI is a mirror for human bias*” (Socher (2019)) since AI models are inherently fed with datasets collected online that contain biased opinions, stereotypes and trends, which are further replicated by the model throughout its training and learning process. One could illustrate this argument with the use of AI in the banking sector, where institutions use AI systems to evaluate an individual’s creditworthiness via credit scoring. Historical data may indicate that the loans granted in the past by the bank have excluded certain minorities, which will further be replicated in the future AI-automated decision-making if AI is used to determine credit scoring. However, minority by itself should not be a determinant of creditworthiness (Socher (2019); Heaven (2021); Andrews (2021)).

Yet AI capabilities more broadly, are increasingly integrated in online and offline products and/or services. Social media, used by approximately 60% of the world’s population (Statista (2022)), relies on AI recommendation systems that have already been subject to scandals in the past (e.g. Cambridge Analytica’s political micro-targeting or Instagram’s toxic recommendation systems encouraging eating disorders among teenage girls, denounced in the Facebook Files). Additionally, other aspects of everyday life such as finance (i.e. loans, as previously mentioned) or even policing (e.g. algorithmic decision-making) are increasingly governed by AI. While applications of AI continuously grow, this trend does not present a major issue in itself. In fact, the primary concern lies in our lack of understanding of the technology, due to the “*black-box*” nature of algorithms (Ghassemi et al. (2021)). Despite advanced research, explainable AI techniques have resulted in “*false hopes*” according to academics (ibid), and the current impossibility to explain AI undermines our control of the technology. Moreover, cases in the past have also highlighted that the technology can be leveraged for malicious uses by ill-intentioned actors (Brundage (2018)).

3 What is the AI Act ?

Given the complexity of the technology and its uses, EU policymakers have drafted a regulation that sets out a legal framework for the “*development, marketing and use of artificial intelligence*” (Commission (2021)), to balance the numerous risks and benefits of AI (Pingen (2023)). The AIA is to apply to all industries, across the whole EU market. To enhance AI trustworthiness, the regulation ensures a ‘product safety regime’ (Kop (2021)) that consists of a risk-based classification system. Each AI system is evaluated individually, and its risk level can be categorized in any of the four tiers : unacceptable, high, limited and minimal (Kop (2021)).

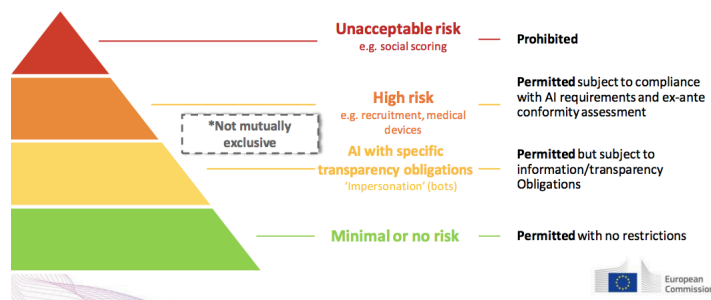


Figure 1: Risk-based approach to regulation, Sioli (2021)

While minimal and limited risk AI systems will have minimal obligations, “*High-Risk*” AI systems will need to undergo a mandatory CE-marking procedure before placed on the market under the current AIA. In order to receive the CE mark, AI providers will need to self-assess, conduct a number of tasks (e.g. registering the AI system on a database, confirming the use of an appropriate quality management system, agreeing to record the system’s technical documentation, etc.) and confirm that the AI system is compliant with the EU’s conformity guidelines (Commission (2021); Edwards (2022)). The original legislation dictates that only a listed subset of niche high-risk AI systems must contract external bodies for the conformity assessment. After the product or services using AI systems are placed on the market, consistent monitoring by the AI providers and their users is requested by the AIA, with the provider being legally required to notify the Market Surveillance Authorities (MSA) – part of the European Commission (EC), if new risks or malfunctioning occur or are reported (Edwards (2022)). For instance, algorithmic creditworthiness assessment systems have been categorized as High-Risk, among others (Gallo (2021)). “*Unacceptable Risk*” applications however, which include manipulation, social scoring and other examples, are to be banned within the EU territory under the current legislative piece (Kop (2021)). However, it is worth mentioning that all companies using AI systems will be subject to continuous assessments, as they will likely develop or integrate new tools over time that may change their risk categorization. If AI providers fail to comply with EU regulation, the AIA reserves penalties for non-compliance (Article 71) which may amount up to €30 million or 6 percent of the company’s worldwide annual turnover, whichever is highest (Commission (2021)). Finally, beyond risk mitigation, the AIA aims to promote investment and innovation (Council of the European Union (2022b); Council of the European Union (2022a)). In addition to other related AI initiatives (e.g. Coordinated Plan on Artificial Intelligence; Council of the European Union (2022a)), the EU’s AIA specifically proposes the creation of “*AI regulatory sandboxes*”. This initiative, to be adopted across all members states, is supposed to loosen the regulatory framework to incentivize innovation in the field of AI, mostly prioritizing small and medium-sized enterprises (Hilliard (2022)).

4 An ongoing European initiative

The AIA marks the first, most stringent and most comprehensive AI regulatory regime of the world’s major institutions (Siegmann & Anderljung (2022)), which inherits the same foundational approach as with the General Data Protection Regulation (GDPR) that emphasises on the “*protection of human dignity and fundamental rights*” (Floridi (2021)). Already back in July 2019, close after Ursula von Der Leyen started her Presidency at the EC, the short-term plan released by the institution comprised a focus on “*Shaping Europe’s digital future*” by paving the way towards the development of cutting-edge and responsive Artificial Intelligence (Commission (2020)). Due to COVID-19, the first proposal of the AIA was only released on April 21st of 2021 by the EC (Floridi (2021)), institution in charge of drawing up proposals for new European legislation and implementing them after the European Parliament’s and Council of the EU’s approval (Union (n.d.b)). On December 6th 2022, the Council of the European Union – made up of EU government ministers who meet to discuss, amend

and adopt laws and usually represent countries (Union (n.d.a)) – finally confirmed the proposal for the AIA. In its position, the Council proposed multiple amendments, such as narrowing down the scope of the AIA. In fact, the definition of an AI system was refined to “*models developed through logic- and knowledge-based systems*”, to draw a clear difference with simpler software (Pingen (2023)) given the ever-evolving characteristics of the technology. Moreover, in line with its core role to represent the interests of member governments, the Council has clarified that the AIA should contain provisions for law enforcement authorities, which has excluded compliance with the AIA for some actors in national security, defense and military purposes (Pingen (2023)). Following the Council’s approval in December, multiple meetings amongst members of the European Parliament (MEPs) Committees from the Internal Market and Consumer Protection (IMCO) and Civil Liberties, Justice and Home Affairs (LIBE) have occurred thereafter to reach a final version proposed for the first vote on May 11th, which was greenlighted by both parties with an overwhelming majority of 84 votes in favour, in comparison to 7 against and 12 abstentions (Sasdeli (2023); Parliament (2023)). While MEPs have proposed substantial amendments to the last version of the approved text, especially in matters of further restrictions for intrusive and discriminatory uses of AI, more controversial points relating to biometric identification technologies or appropriate policy responses for the latest rise of generative AI (e.g. ChatGPT) are planned to be discussed (Sasdeli (2023)). As a next step, the AI Act will undergo a plenary vote by the European Parliament scheduled in June 2023, which will be followed by “*trilogue*” negotiations, involving final amendments between the three main decision-making bodies : the Council of the European Union, the Commission and the Parliament (Lovells (2023)). These trilateral meetings are usually informal and reunite representatives from each institutional body, with the goal to reach an agreement on the given piece of legislation (Europa, n.d.).

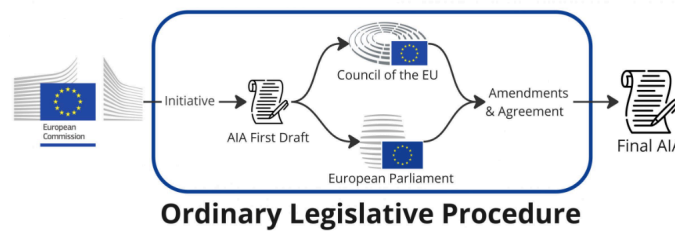


Figure 2: Ordinary Legislative Procedure, Artificial Intelligence Act (2022)

5 AI Act controversies

While the AI Act has been praised by multiple experts in AI regulation such as professor S.Wachter, who has highlighted the benefits of a “*harmonized system*” (Hsu (2022)) that could apply to the EU and beyond, critics have pointed out the AI act’s flawed approach to regulation (Watch (2021)). In fact, slow policy responses to a changing environment may jeopardise the AIA’s effectiveness within the EU and outside of it. Firstly, it is important to note that the AIA is the outcome of a three-year long process in which the EC has set up an expert group and public consultations on the best AI regulatory approach (Covington (2022)); yet newer generative AI (G-AI) or

“foundation models” that have emerged in the last year may not be captured effectively by the legislation piece. G-AI’s stakes and implications may be even greater, because of the capability of AI systems to generate new outputs based on training data (Routley (2023)), which marks a radical change with traditional categories of AI that were initially targeted by policymakers. Therefore, these will remain an important topic of discussion in the future meetings as the technology evolves and policymakers gain understanding. GPT-4, known as one of the latest most powerful G-AI conversational chatbots released by OpenAI, has already demonstrated *“sparks of AGI”* according to researchers (Bubeck et al. (2023)), which could pose multiple threats to society. Researchers of the AI system have demonstrated in an experiment, how GPT-4 could *“pretend to be blind to trick a human TaskRabbit Worker into solving a CAPTCHA”* (Hurler (2023); OpenAI (2023)), and other examples demonstrate how G-AI systems pose concerns for trust and security, due to cases of system hallucination, deepfakes, data privacy leakages, copyrights issues, and cybersecurity problems (Ortiz (2023)).

While the lead lawmakers Benifei and Tudorache have ruled out to add G-AI systems in the high-risk category in the latest amendments of the text, which has consequently been approved by the Parliament (Volpicelli (2023)), the obligations imposed on foundation models’ providers (e.g. robust protection of fundamental right, assessment and risk mitigation of models, registration of the models in the EU database as for other AI models; Lovells (2023)) may lack efficiency according to critics. In fact, Hacker et al. (2023) argue that the AI Act may inadequately address G-AI risks as the overly burdensome legislation focuses on regulating the pre-trained model (i.e. the technology) or its characteristics, rather than the applications that can be developed with it. However, as down-stream G-AI use cases are developed and companies integrate them in their processes through the means of APIs or plug-ins for example, regulators should enforce an output-based regulatory approach (Goldman (2022)). For instance, building ‘perfect’ models that abide by the regulatory requirements of the EU does not necessarily make them ‘safe’ for use when in malicious hands, who could use them for ill-intentioned purposes (Goldman (2022)). Therefore, it could be argued that G-AI research needs to continue in an open environment, without strict constraints, to be able to control the technology and use it for beneficial applications. Leading research institutes, scientists and business representatives have called for the legislative draft to be amended in an open letter to the Parliament that considers this approach (Hahn (2023)). The letter recommends to *“impose requirements proportional to risk”*, or according to the intended applications (LAION (2023)) as the current new requirements for foundational models could slow down innovation and jeopardize Europe’s future. While further fine-tuned versions of the AIA addressing more aspects of foundation models have been proposed right before the vote on May 11th, day on which the latest version has been approved, the amendments do not speak to the concerns of aforementioned output regulation. The EU plans to further discuss the subject in the future trilateral meetings (Sasdeli (2023)).

6 Future of the AI act : towards a Brussels effect ?

Some have argued that a possible resolution could be expected before the end of 2023, allowing the world’ first AI legislation to enter into force in the next 2-3 years, given

the “*vacatio legis*” timeframe to adapt (Von Gerlach & Whitehead (2023); Lukianets (2021)). Until the resolution, controversial debates revolving around the AIA’s scope and implementation will continue between EU bodies and players of the tech ecosystem. Nonetheless, Engler (2022) has already claimed that the proposal was “sufficiently settled”, to analyse its likely international impact. While it is important to highlight that the AIA will automatically have an extraterritorial impact, since companies operating in the European geography will need to comply with the new set of rules, the potential for a “*Brussels effect*” remains unclear (Engler (2022)). The term refers to a global diffusion of an EU’s standards or piece of legislation, which can both have a “*de facto*” effect, materialized by a compliance with EU regulation in non-EU countries, and a “*de jure*” effect, where other jurisdictions are influenced by EU’s trustworthy AI values and adopt similar regulatory provisions (Siegmann & Anderljung (2022), p.2, p.3). The likelihood of a Brussels effect for the AIA would therefore depend on the magnitude of those two parameters. Siegmann & Anderljung (2022)p.2, p.5, have argued in favour of a significant AIA Brussels effect, by offering findings for likely effects for both *de facto* and *de jure*. In fact, from a *de facto* perspective, it would be expected that EU-compliant AI products are commercialized outside of the EU, given the “*favourable market properties*” of the EU, namely the regulatory sandboxes that support and push efforts towards responsible AI Research and Development that can be applied globally. This is further reinforced as the “*inelastic demand*” for AIA-targeted products incentivizes companies to operate in the EU despite the regulatory compliance (Siegmann & Anderljung (2022)). The authors have also pointed out “high differentiation costs” for creating separate compliant and non-compliant versions of a same product, which may encourage companies to consolidate processes and offer EU-compliant products globally, strengthening the *de facto* effect. To illustrate this with an analogous example, Chen et al. (2022) have highlighted the expensive compliance costs incurred by companies to abide by the EU’s GDPR regulation, roughly estimated at \$3million for SMEs and \$16 million for Fortune 500 firms in 2017-2018 (Prasad (2020)). Following on this example, evidence by Batikas et al. (2020) has shown that websites and web technology providers catering to non-EU audiences (not subject by GDPR) would still comply with the regulation, an effect that could likely happen with the AIA. Especially in terms of the emerging foundation models, developers may wish to ensure that their systems are AI-compliant from the beginning due to the high Research and Development costs, which may strengthen the effect for the AIA (Siegmann & Anderljung (2022)).

Moreover, *de jure* effects have already followed although the Brussels effect had not yet been officially adopted, whether on the national level or international level. Italy’s temporary ban on ChatGPT by the national data protection authority Garante in March, could be considered as a national Brussels effect based on the EU’s style of ex-ante prevention policymaking. Italy assessed the systems as high risk, dictated the ban and only lifted it once OpenAI added sufficient AI safety measures for children, such as an age verification and parental consent process (IAPP (2023)). At an international level, Brazil has also presented a draft proposition for regulating AI, which is based on similar risk classification characteristics as the AI act, likely to result from an early-stage Brussels effect (Kohn (2023)). China, Japan, India, and Canada among others, are also on the path of exploring AI regulation possibilities (Kohn (2023)).

The AIA will likely have significant impacts on their choices as countries may wish to maintain win-win harmonized standards given the aforementioned cost burdens. Conversely, those who do not believe in an as strong Brussels effect have criticized the EU's over-regulatory regime on AI, as the actual benefits of the AIA will be difficult to measure (Engler (2022)). Ultimately, this may lead to a lower perceived value in regulatory competition and act as a disincentive for other countries to adopt similar legislation (de jure effect; Engler (2022)). Moreover, it could be argued that the countries following on the EU's path may do so independently, given the absolute need to regulate AI and newer G-AI systems (i.e. "*Pause Giant Experiments*" open letter). As a consequence, the true effects attributed to a "*Brussels effect*" may be harder to measure, in the case of the AI Act.

7 Conclusion

This paper argues that as AI evolves and is integrated in new products and services, a regulation that sets specific guardrails for the technology is needed. The AI Act may become the world's first and most stringent AI regulation, but requires a better understanding and coordination with the private sector to offer a relevant framework that ensures trustworthy AI. While the EU's final version of the AIA is yet to be decided, a partial Brussels effect may occur despite measurement difficulties.

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